

Sign in / Sign up
Return BioPublisher | Journal List
Feedback | Profile | Support

[Home](#)[Articles](#)[Search](#)[My MPR](#)[About MPR](#)

Research Insight

The Interaction Between *Ganoderma lucidum* Polysaccharides and Gut Microbiota: Implications for Immune Health [Open Access](#)

Zhichao Chen ¹ , Haijuan Zhang ² 

1 Department of Traditional Chinese Medicine, Longquan People's Hospital, Longquan, Zhejiang, 323000, China

2 Longquan Agricultural and Rural Bureau, Longquan, Zhejiang, 323700, China

 Author  Correspondence author

Medicinal Plant Research, 2025, Vol. 15, No. 1 doi: 10.5376/mpr.2025.15.0001

Received: 08 Dec., 2024 Accepted: 10 Jan., 2025 Published: 21 Jan., 2025

© 2025 BioPublisher Publishing Platform

This is an open access article published under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

Preferred citation for this article:

Chen Z.C., and Zhang H.J., 2025, The interaction between *Ganoderma lucidum* polysaccharides and gut microbiota: implications for immune health, Medicinal Plant Research, 15(1): 1-11 (doi: [10.5376/mpr.2025.15.0001](https://doi.org/10.5376/mpr.2025.15.0001))

Abstract

Ganoderma lucidum polysaccharides (GLPs) are natural polysaccharide compounds extracted from *Ganoderma lucidum*, known for their diverse biological activities. This study explores the interaction between GLPs and gut microbiota and its effects on immune health, revealing how GLPs regulate immune responses through gut microbiota and investigating their potential applications in preventing and treating immune-related diseases. The study shows that GLPs can increase the production of short-chain fatty acids, regulate the proportion of beneficial gut bacteria, and alleviate inflammatory responses by modulating the TLR4/MyD88/NF-κB signaling pathway. The results indicate that GLPs significantly improve gut microbiota imbalance, enhance macrophage phagocytosis, and increase the cytotoxicity of natural killer cells, demonstrating their potential in immune regulation and disease prevention. Future research will focus on further uncovering the molecular mechanisms of GLPs and their long-term application potential in human health.

Keywords

Ganoderma lucidum polysaccharides; Gut microbiota; Immune health; Interaction; Disease prevention

1 Introduction

Ganoderma lucidum is a medicinal mushroom with a long history of use in traditional Chinese medicine. The polysaccharides derived from *G. lucidum* (GLPs) are notable for their diverse biological activities, including antioxidant, antitumor, anti-inflammatory, antiviral, anti-diabetes, and immunomodulatory effects ([Wachtel-Galor et al., 2011](#); [Lu et al., 2020](#)). These polysaccharides are characterized by their complex molecular structures, which include variations in molecular weight, branching, and monosaccharide composition ([Lu et al., 2020](#)). The gut microbiota, a complex community of microorganisms residing in the gastrointestinal tract, plays a crucial role in maintaining host health. It is involved in various physiological processes, including digestion, metabolism, and the modulation of the immune system. The composition and function of the gut microbiota can be influenced by diet, lifestyle, and the intake of bioactive compounds such as polysaccharides ([Kubota et al., 2018](#); [Li et al., 2020](#)).

The gut microbiota is integral to the development and function of the host's immune system. It contributes to the maturation of immune cells, the production of antimicrobial peptides, and the regulation of inflammatory responses ([Kubota et al., 2018](#)). The interaction between the gut microbiota and the immune system is bidirectional; while the microbiota influences immune function, the immune system also shapes the composition of the gut microbiota. Disruptions in this delicate balance can lead to immune-related disorders, including allergies, autoimmune diseases, and infections ([Li et al., 2020](#)).

Understanding the interaction between GLPs and gut microbiota is of great significance. Research has shown that GLPs can regulate immune responses, potentially mediated through their effects on the gut microbiota ([Chen et al., 2019](#)). This regulatory effect can improve gut barrier function, reduce inflammatory responses, and enhance immune function ([Jin et al., 2017](#); [Guo et al., 2021](#)). For example, GLPs can increase the production of immunoglobulin A (IgA) and antimicrobial peptides in the gut, strengthening the gut barrier and reducing the risk of infection ([Kubota et al., 2018](#)). Additionally, the effects of GLPs on the gut microbiota can have systemic impacts, potentially offering therapeutic benefits for diseases such as obesity, diabetes, and cancer ([Chang et al., 2015](#); [Sang et al., 2021](#)). These interactions may have profound implications for developing novel therapeutic strategies aimed at enhancing immune health through dietary interventions.

This study explores the interaction between *Ganoderma lucidum* polysaccharides (GLPs) and gut microbiota, and their effects on immune health, particularly focusing on how GLPs regulate the composition and function of gut microbiota, identifying microbial metabolites involved in immune modulation, and assessing their overall impact on immune function. The study

Medicinal Plant Research

• Volume 15



View Options

- PDF(1078KB)
- FPDF(win)
- FPDF(mac)
- HTML
- Online fPDF

Associated material

- Readers' comments

Other articles by authors

- Zhichao Chen 
- Haijuan Zhang 

Related articles

- *Ganoderma lucidum* polysaccharides 
- Gut microbiota 
- Immune health 
- Interaction 
- Disease prevention 

Tools

- Email to a friend
- Post a comment

anticipates that GLPs will promote the growth of beneficial gut bacteria, improve gut barrier integrity, and modulate immune responses, thereby offering potential therapeutic strategies for immune-related diseases. This study attempts to provide new insights into the role of dietary polysaccharides in regulating the gut microbiota and supporting immune health, while also offering theoretical foundations and references for the development of disease interventions based on GLPs.

2 Chemical Composition and Biological Activities of *Ganoderma lucidum* Polysaccharides

2.1 Structural characteristics

Ganoderma lucidum polysaccharides (GLPs) are composed of various monosaccharides, including glucose, galactose, and mannose. Their specific composition varies depending on the extraction method and the part of the *G. lucidum* used. For instance, GLP-1 and GLP-2 are purified from *G. lucidum* via gradient ethanol precipitation and anion-exchange chromatography, respectively (Li et al., 2019). Structural characterization has shown that GLP-1 is a D-galactoglucan containing $\rightarrow 6$ - β -D-glucose (GlcP)- and $\rightarrow 6$ - α -D-galactose (GalP)- residues, with a random linear conformation. In contrast, GLP-2 is a spherical β -D-glucan composed of $\rightarrow 6$ - β -D-GlcP and $\rightarrow 3$ - β -D-GlcP residues.

The molecular weight of *Ganoderma lucidum* polysaccharides (GLPs) ranges from several thousand to millions, which significantly affects their biological functions. GLPs exhibit various structural features, including both linear and branched conformations. For example, GLP-1, with a flexible random linear conformation, demonstrates superior immunomodulatory activity, effectively protecting the spleen and thymus, promoting hematopoiesis, and increasing serum IgA levels (Li et al., 2019). On the other hand, GLP-3, extracted from *Ganoderma lucidum*, is a spherical polysaccharide with immunomodulatory properties. Its main chain is composed of α -D-glucose, and research has shown that GLP-3 enhances immune regulation by activating specific signaling pathways, which is related to its tertiary helical structure (Gao et al., 2020).

2.2 Immunomodulatory effects

Studies have found that *Ganoderma lucidum* polysaccharides (GLPs) can activate innate immune cells. For example, GLPs can regulate macrophage polarization through the MAPK/NF- κ B signaling pathway, enhancing the expression of the M1 phenotype (pro-inflammatory) and inhibiting the M2 phenotype (anti-inflammatory), thereby suppressing tumor cell growth (Li et al., 2018; 2023). Additionally, GLPs can modulate the gut microbiota by inhibiting the TLR4/MyD88/NF- κ B signaling pathway, improving intestinal barrier function, reducing the expression of inflammatory markers, and thus alleviating colonic inflammation and tumorigenesis (Guo et al., 2021).

GLPs can modulate the production of various cytokines, which are signaling molecules that mediate and regulate immunity, inflammation, and hematopoiesis. For example, GLP treatment has been shown to downregulate pro-inflammatory cytokines such as IL-1 β , iNOS, and COX-2, thereby reducing inflammation (Guo et al., 2021). And GLPs can enhance the production of anti-inflammatory cytokines, contributing to their overall immunomodulatory effects (Li et al., 2019).

2.3 Extraction and purification techniques

Hot water extraction (HWE) is one of the most commonly used methods for extracting *Ganoderma lucidum* polysaccharides. This method involves boiling *G. lucidum* materials in water to dissolve the polysaccharides. The extract is then concentrated and purified using techniques such as ethanol precipitation and ion-exchange chromatography (Zheng et al., 2022). This method is highly effective in yielding polysaccharides with significant biological activity. One study found that polysaccharides extracted using HWE had a relatively large molecular weight, reaching 703.45 kDa (Kang et al., 2019).

Enzymatic extraction, on the other hand, utilizes specific enzymes to break down the cell walls of *G. lucidum*, releasing the polysaccharides. Compared to hot water extraction, this method may be more efficient and selective. A study comparing different extraction methods-including hot water extraction, ultrasound-assisted extraction, and enzyme-assisted extraction-showed that enzyme-assisted extraction produced polysaccharides with higher content and antioxidant activity, outperforming traditional hot water extraction, particularly in terms of immunomodulatory effects (Liu et al., 2021). For example, polysaccharides extracted through enzyme-assisted methods significantly enhanced macrophage proliferation and cytokine secretion, especially promoting the production of NO, TNF- α , and IL-6.

3 Gut Microbiota and Its Role in Human Health

3.1 Composition and diversity of gut microbiota

The human gut microbiota is a complex ecosystem primarily composed of bacteria, with the major phyla being Firmicutes, Bacteroidetes, Actinobacteria, and Proteobacteria. Specific species within these phyla play crucial roles in maintaining gut health. For instance, beneficial bacteria such as *Bacteroides* and *Blautia* are known to support metabolic functions and immune health, while harmful bacteria like *Aerococcus* and *Proteus* can contribute to disease states (Chen et al., 2019; Lv et al., 2019).

Gut microbiota perform essential functions, including the fermentation of indigestible carbohydrates to produce short-chain fatty acids (SCFAs), which are vital for colon health and energy metabolism. They also play a role in synthesizing vitamins, metabolizing bile acids, and modulating the immune system. The metabolic activities of gut microbiota are influenced by their composition, which can be altered by dietary components such as polysaccharides from *Ganoderma lucidum* (Chen et al., 2019; Guo et al., 2020; Guo et al., 2021).

3.2 Factors influencing gut microbiota balance

Diet is a primary factor influencing the composition and function of gut microbiota. High-fat diets can lead to dysbiosis, characterized by a decrease in beneficial bacteria and an increase in harmful species. Conversely, dietary polysaccharides from *Ganoderma lucidum* have been shown to modulate gut microbiota positively, increasing the abundance of beneficial bacteria like *Bacteroides* and reducing harmful bacteria such as *Ruminococcus* (Lv et al., 2019; Sang et al., 2021; Zheng et al., 2022).

Antibiotics can severely disrupt the gut microbiota, leading to decreased microbial diversity, dysbiosis, and potentially impairing gut barrier function and immune responses. However, combining antibiotics with prebiotics such as *Ganoderma*

lucidum polysaccharides can mitigate these adverse effects by promoting the growth of beneficial bacteria and enhancing the integrity of the gut barrier (Li et al., 2023). For example, studies have found that the use of ciprofloxacin (an antibiotic) alone damages the gut barrier, while GLPs alleviate the negative effects of antibiotics on the gut by increasing the concentration of tight junction proteins such as ZO-1 and Occludin.

3.3 Interaction between gut microbiota and immune system

The gut microbiota plays a crucial role in the development and function of the immune system, promoting the maturation of immune cells and contributing to the production of immunoregulatory compounds. Gut microbiota dysbiosis is linked to various immune disorders, including inflammatory bowel disease (IBD) and metabolic syndrome (Tong et al., 2019). Studies have shown that GLPs can enhance immune function by modulating the gut microbiota, thereby increasing the activity of macrophages and the cytotoxicity of natural killer (NK) cells (Wu et al., 2020; Su et al., 2021).

Further research by Guo et al. (2021) demonstrated that GLPs can improve gut microbiota imbalance, increase the production of short-chain fatty acids, and alleviate endotoxemia. And GLPs reduce the risk of immune-related diseases by inhibiting inflammation-related signaling pathways, such as the TLR4/MyD88/NF- κ B pathways (Tong et al., 2019; Guo et al., 2021). Moreover, broken-spore polysaccharides from *Ganoderma lucidum* have been shown to effectively inhibit obesity, inflammation, and fat accumulation induced by a high-fat diet by modulating the gut microbiota and improving gut barrier function (Sang et al., 2021).

4 Interaction Between *G. lucidum* Polysaccharides and Gut Microbiota

4.1 Modulation of gut microbiota by *G. lucidum* polysaccharides

Ganoderma lucidum polysaccharides (GLPs) have been shown to significantly modulate the composition of gut microbiota. Studies have demonstrated that GLPs can restore the balance of gut microbiota in various disease models. For instance, GLPs were found to reduce the abundance of harmful bacteria such as *Aerococcus*, *Ruminococcus*, *Corynebacterium*, and *Proteus*, while increasing beneficial bacteria like *Blautia*, *Dehalobacterium*, *Parabacteroides*, and *Bacteroides* in type 2 diabetic rats (Chen et al., 2019).

A study on rats fed a high-fat diet revealed that GLPs can alleviate lipid metabolism disorders and impaired gut barrier function caused by the high-fat diet by modulating the gut microbiota, such as increasing the abundance of beneficial bacteria like *Prevotella* (Tong et al., 2019). In a colon cancer model, GLPs significantly improved symptoms of colitis and colon cancer by reducing the expression of inflammatory factors and modulating the gut microbiota, such as decreasing cancer-associated bacteria like *Oscillospira*. These findings highlight the potential of GLPs in cancer prevention and treatment (Luo et al., 2018). The results suggest that GLPs may serve as potential prebiotics, promoting the growth of beneficial gut microbes while inhibiting harmful bacteria.

4.2 Mechanisms of interaction at molecular level

The interaction between GLPs and gut microbiota involves several molecular mechanisms. GLPs have been shown to influence the gut microbiota through their structural characteristics, such as molecular weight, monosaccharide composition, and glycosidic bonds (Li et al., 2019; Lu et al., 2020). These structural features determine the physicochemical properties and biological activities of GLPs, which in turn affect their interaction with gut microbiota.

For example, GLPs can bind to receptors on immune cells, such as dectin-1 on macrophages, monocytes, dendritic cells, and neutrophils, leading to the activation of signaling pathways like MAPKs and NF- κ B, which are involved in cytokine production and immune response (Ahmad et al., 2021). The results indicate that GLPs can enhance the activity of T cells and macrophages, increasing the production of important cytokines such as IL-2 and TNF- α , thereby boosting the overall immune response (Figure 1). Furthermore, GLPs have been shown to modulate the gut microbiota by enhancing gut barrier function, increasing the number of goblet cells, and promoting the secretion of mucins and tight junction proteins (Guo et al., 2021).

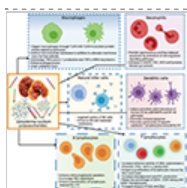


Figure 1 *G. lucidum* polysaccharides produce immunomodulatory action through promoting antigen-presenting cells, mononuclear phagocyte system, humoral and cellular immunity (Adopted from Ahmad et al., 2021)

Image Caption: The figure illustrates how *Ganoderma lucidum* polysaccharides (GLPs) produce immunomodulatory effects by promoting antigen-presenting cells, the mononuclear phagocyte system, humoral, and cellular immunity. The figure details the mechanism of GLPs in activating T cells and increasing cytokine production, such as IL-2 and TNF- α . The results indicate that GLPs can enhance the body's immune response through multiple immune pathways, showing significant potential, particularly in antiviral and anti-infection applications (Adapted from Ahmad et al., 2021)

4.3 Influence on microbial metabolites

GLPs not only modulate the composition of gut microbiota but also influence the production of microbial metabolites. For instance, GLPs have been reported to increase the production of short-chain fatty acids (SCFAs), which are beneficial metabolites produced by gut bacteria through the fermentation of dietary fibers (Guo et al., 2021). SCFAs play a crucial role in maintaining gut health by providing energy to colonocytes, regulating immune responses, and protecting against inflammation and tumorigenesis.

GLPs have been shown to restore disturbed amino acid, carbohydrate, and nucleic acid metabolism in the gut microbiota of type 2 diabetic rats, indicating their potential to normalize metabolic functions (Chen et al., 2019). Moreover, GLPs have been found to regulate the levels of key metabolites such as dopamine, prolyl-glutamine, and L-threonine, which are

associated with immune enhancement and overall health (Wu et al., 2020). These findings highlight the significant impact of GLPs on microbial metabolites and their potential therapeutic benefits.

5 Case Studies

5.1 Study on the inhibitory effects of *G. lucidum* polysaccharides on colitis-associated cancer

Colorectal cancer (CRC) is the third leading cause of cancer-related deaths worldwide. Its occurrence is associated with multiple factors, such as environmental, dietary, genetic, and epigenetic influences (Kendong et al., 2021). Gut microbiota dysbiosis is considered a key factor in inducing chronic inflammation and promoting CRC development. In recent years, research on the prevention of CRC through prebiotics and dietary fibers, which modulate gut microbiota, has been increasing. *G. lucidum* polysaccharides (GLPs), as a significant bioactive substance, have garnered attention for their immunoregulatory and anticancer properties (Chattopadhyay et al., 2021; Guo et al., 2021).

Guo et al. (2021) explored the effects of GLPs on an AOM/DSS-induced mouse model of colitis and tumorigenesis. The study found that GLP increased the production of short-chain fatty acids (SCFAs), inhibited the TLR4/MyD88/NF- κ B signaling pathway, and reduced macrophage infiltration and the expression of inflammatory markers (Figure 2). Additionally, GLP showed the ability to suppress LPS-induced inflammatory markers in in vitro experiments. The results suggest that GLP can significantly reduce tumor size and number, improve inflammatory responses, and enhance intestinal barrier function by modulating gut microbiota, providing a new research direction for CRC prevention and treatment.



Figure 2 Schematic showing the role of GLP in modulating the intestinal microbiota and its relationship with inflammation and tumorigenesis in AOM/DSS mice (Adopted from Guo et al., 2021)

Image caption: The figure reveals the mechanism by which GLP modulates the structure of the gut microbiota, increases the production of short-chain fatty acids (SCFAs), and improves intestinal barrier function, thereby inhibiting inflammation and tumorigenesis. GLP suppresses the TLR4/MyD88/NF- κ B and MAPK (ERK and JNK) signaling pathways, reducing macrophage infiltration and pro-inflammatory cytokine expression, ultimately lowering the risk of colitis and tumor formation (Adapted from Guo et al., 2021)

5.2 Study on the immunomodulatory effects of sporoderm-broken *G. lucidum* spore polysaccharides

G. lucidum spores are the reproductive cells of the fungus, and their outer walls are relatively hard, usually requiring sporoderm-breaking treatment to extract the active components. Both *G. lucidum* polysaccharides and *G. lucidum* spore polysaccharides are polysaccharides in chemical structure, but due to differences in their extraction sources, their molecular weight, glycan chain structure, and accompanying bioactive compounds may differ, leading to variations in their biological activities (Wu et al., 2020). Crude polysaccharides and refined polysaccharides from *G. lucidum* spores (CPGS and RPGS) are specific types of *G. lucidum* spore polysaccharides, belonging to the *G. lucidum* (GLPs) family. They differ from traditional GLPs in terms of source, composition, and bioactivity. After sporoderm-breaking, the polysaccharide components of *G. lucidum* spores exhibit higher bioavailability (Sang et al., 2021; Su et al., 2021). Compared to GLPs, CPGS and RPGS have distinct advantages in immune regulation and gut microbiota modulation, especially RPGS, which, due to its higher purity, demonstrates stronger immunostimulatory effects (Su et al., 2021).

The study by Su et al. (2021) evaluated the immunomodulatory effects of CPGS and RPGS on the immune system in mice. The results showed that CPGS significantly enhanced the tumor-killing ability of natural killer cells, while RPGS notably increased the proportion of T cells and their subsets in peripheral blood and promoted T cell activation. Additionally, 16S rRNA sequencing analysis of the gut microbiota revealed that CPGS and RPGS reshaped the gut microbiome, enriching polysaccharide-metabolizing bacteria such as *Adlercreutzia* and *Prevotella*, which were closely related to T cell activity. The findings indicate that *G. lucidum* spore polysaccharides can significantly enhance the function of the adaptive immune system by regulating the gut microbiota and immune cell activity, further supporting their potential application in promoting health.

5.3 Prebiotic effects of *G. lucidum* polysaccharides (GLPs) in a human intestinal microbial ecosystem

In recent years, the prebiotic potential of *G. lucidum* polysaccharides (GLPs) has garnered increasing attention. Prebiotics can promote the growth of beneficial gut microbes, regulate the host's intestinal environment, and provide health benefits (Guo et al., 2021; Sang et al., 2021). To better understand the metabolic and microbial regulatory effects of GLPs in the gut, a study employed the Simulator of the Human Intestinal Microbial Ecosystem (SHIME) to evaluate the impact of GLPs on human fecal microbiota. The results showed that GLPs served as a carbon source for gut microbiota, significantly increasing the production of short-chain fatty acids (such as acetic, propionic, and butyric acids) and lowering intestinal pH, thus inhibiting the growth of potential pathogens (Yang et al., 2022). GLPs also modulated the gut microbiota, notably enhancing the abundance of Bacteroidetes, particularly enriching *Bacteroides ovatus* and *B. uniformis*, which are closely associated with improved metabolic function and enhanced immune regulation (Figure 3).

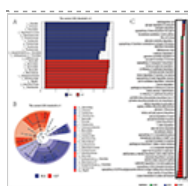


Figure 3 LefSe analysis and PICRUST2 analysis of microbiota between the GLP group and the BLK group at ASV level. (A) LDA scores with corresponding phylum, genus and species (LDA score>4.0 was shown) and (B) Cladogram of the

bacteria taxa. (C) LDA scores of differential metabolic pathways. BLK, blank group; GLP, polysaccharides from fruiting bodies of *Ganoderma lucidum* (Adopted from [Yang](#) et al., 2022)

Image caption: The figure shows the impact of *G. lucidum* polysaccharides (GLPs) on gut microbial metabolic functions and the abundance of specific bacterial taxa. Figures A and B, analyzed by LEfSe, demonstrate that the GLP group significantly enriched Bacteroidetes and Proteobacteria, while the control group showed enrichment of Firmicutes and Fusobacteria. Figure C presents the PICRUST2 functional analysis, indicating significant differences in metabolic pathways, such as glycogen degradation and secondary metabolite synthesis, between the GLP group and the control group. These results reveal that GLPs not only alter microbial composition but also significantly affect their metabolic functions, confirming their potential in regulating gut metabolic health (Adapted from [Yang](#) et al., 2022)

The study suggests that GLPs, as a prebiotic, has potential anti-inflammatory and metabolic health benefits by regulating gut microbiota and their metabolites, providing scientific support for its further application in gut health.

6 Potential Applications in Functional Foods and Medicine

6.1 Use in functional foods for gut health improvement

Ganoderma lucidum polysaccharides (GLPs) have shown significant potential as functional food ingredients aimed at improving gut health. Studies have demonstrated that GLPs can modulate gut microbiota composition, enhancing the abundance of beneficial bacteria such as *Bacteroides*, *Parabacteroides*, and *Lactobacillus*, while reducing harmful bacteria like *Aerococcus* and *Ruminococcus* ([Chang](#) et al., 2015; [Chen](#) et al., 2019; [Zheng](#) et al., 2022). This modulation leads to improved gut barrier function, increased production of short-chain fatty acids (SCFAs), and reduced endotoxemia, which collectively contribute to better gut health ([Guo](#) et al., 2021; [Sang](#) et al., 2021; [Li](#) et al., 2023). Additionally, GLPs has been found to alleviate gut microbiota dysbiosis induced by high-fat diets and antibiotics, further supporting its role as a prebiotic agent in functional foods ([Sang](#) et al., 2021; [Li](#) et al., 2023).

6.2 Pharmaceutical applications for immune health

The immunomodulatory properties of GLP make it a promising candidate for pharmaceutical applications aimed at enhancing immune health. Research has shown that GLP can enhance macrophage phagocytosis, natural killer (NK) cell cytotoxicity, and overall immune cell function ([Wu](#) et al., 2020; [Guo](#) et al., 2021). These effects are mediated through the modulation of gut microbiota, which in turn influences the production of key metabolites involved in immune responses ([Jin](#) et al., 2019; [Wu](#) et al., 2020). Furthermore, GLP has demonstrated anti-inflammatory and anticancer properties by inhibiting pro-inflammatory cytokines and oncogenic signaling pathways, making it a potential therapeutic agent for conditions like colorectal cancer and chronic inflammation ([Khan](#) et al., 2019; [Guo](#) et al., 2021). The ability of GLP to restore gut microbiota balance and improve metabolic profiles in diabetic models also highlights its potential in managing metabolic disorders ([Chen](#) et al., 2019).

6.3 Challenges in commercialization

Despite the promising applications of GLPs in functional foods and pharmaceuticals, several challenges need to be addressed for successful commercialization. One major challenge is the standardization of GLPs extraction and purification processes to ensure consistent quality and bioactivity ([Zheng](#) et al., 2022; [Swallah](#) et al., 2023). Variability in the structural characteristics of GLPs, such as molecular weight and monosaccharide composition, can affect its efficacy and safety, necessitating rigorous quality control measures ([Zheng](#) et al., 2022). Additionally, the scalability of GLPs production and the cost-effectiveness of manufacturing processes are critical factors that need optimization. Regulatory hurdles related to the approval of GLPs as a functional food ingredient or pharmaceutical agent also pose significant challenges. Comprehensive toxicological studies and clinical trials are required to establish the safety and efficacy of GLPs, which can be time-consuming and costly ([Swallah](#) et al., 2023). Addressing these challenges through collaborative research and development efforts will be essential for the successful commercialization of GLPs-based products.

7 Future Research Directions

7.1 Understanding specific gut microbial strains involved

Future research should focus on identifying specific gut microbial strains that are positively influenced by *Ganoderma lucidum* polysaccharides (GLPs). Studies have shown that GLPs can increase the abundance of beneficial bacteria such as *Lactobacillus*, *Turicibacter*, and *Romboutsia*, while reducing harmful bacteria like *Staphylococcus* and *Helicobacter* ([Wu](#) et al., 2020). Additionally, GLPs has been found to enrich bacteria such as *Adlercreutzia*, *Parabacteroides*, and *Prevotella*, which are associated with immune regulation ([Su](#) et al., 2021). Identifying these key strains can help in understanding the precise mechanisms through which GLP exerts its beneficial effects. The role of microbial metabolites in mediating the immune response is another critical area for future research. GLP has been shown to modulate a range of metabolites, including short-chain fatty acids (SCFAs) and other bioactive compounds like dopamine and prolyl-glutamine, which are linked to immune enhancement ([Guo](#) et al., 2021). Understanding how these metabolites interact with the host immune system can provide deeper insights into the immunomodulatory effects of GLPs.

7.2 Long-term effects of *Ganoderma lucidum* polysaccharides consumption

Long-term studies are needed to evaluate the sustained immunomodulatory benefits of GLPs. While short-term studies have demonstrated enhanced macrophage phagocytosis and NK cell cytotoxicity ([Wu](#) et al., 2020), it is essential to investigate whether these benefits persist over extended periods. This will help in determining the long-term efficacy of GLPs as an immune-boosting supplement. The safety and tolerability of long-term GLPs consumption should also be a focus of future research. Although current studies have not reported significant adverse effects, comprehensive long-term studies are necessary to ensure that GLPs is safe for prolonged use ([Jin](#) et al., 2019). This will involve monitoring for any potential side effects and assessing the overall health impact of extended GLP consumption.

7.3 Integration with other prebiotics and probiotics

Investigating the synergistic effects of GLPs when combined with other prebiotics is another promising research direction. GLPs has shown potential as a prebiotic by modulating gut microbiota and improving gut barrier function ([Krumbeck](#) et al., 2018; [Chen](#) et al., 2019). Combining GLPs with other prebiotics could enhance these effects, leading to improved gut health

and immune function. The development of combined probiotic supplements that include GLPs is another area worth exploring. Studies have shown that GLP can increase the abundance of probiotic bacteria such as *Lactobacillus* and *Bifidobacterium* (Lv et al., 2019; Li et al., 2023). Formulating supplements that combine GLP with these probiotics could offer a more comprehensive approach to gut health and immune support.

8 Conclusion

The interaction between *G. lucidum* polysaccharides (GLPs) and gut microbiota has shown significant implications for immune health. Studies have demonstrated that GLPs can modulate gut microbiota composition, enhance immune cell function, and improve gut barrier integrity. For instance, GLPs has been found to decrease inflammation and tumorigenesis in colitis models by modulating gut microbiota and increasing short-chain fatty acid production. And GLPs has been shown to enhance macrophage phagocytosis and NK cell cytotoxicity, correlating with changes in gut microbiota and metabolite profiles. Furthermore, GLPs administration has been linked to improved adaptive immune function and increased proportions of beneficial gut bacteria. These findings collectively suggest that GLPs act as a promising prebiotic with potential therapeutic benefits for immune health.

Despite the promising results, several limitations exist in the current body of research. Most studies have been conducted in animal models, which may not fully replicate human physiological responses. The specific mechanisms by which GLP modulates gut microbiota and immune responses are not yet fully understood, necessitating further molecular and mechanistic studies. Additionally, variations in the extraction methods and dosages of GLP used across different studies may lead to inconsistent results, highlighting the need for standardized protocols. Long-term safety and efficacy studies in humans are required to validate the therapeutic potential of GLP for immune health.

The findings from these studies suggest several potential implications for future health interventions. GLP could be developed as a dietary supplement or functional food ingredient aimed at enhancing immune health and preventing inflammatory diseases. Synergistic effects of GLP with other treatments, such as antibiotics, could be explored to mitigate the adverse effects of conventional therapies and improve patient outcomes. Future research should focus on clinical trials to establish the efficacy and safety of GLP in humans, as well as on elucidating the precise molecular pathways involved in its immunomodulatory effects. At the same time, personalized nutrition plans can be developed based on individual gut microbiota profiles to maximize the health benefits of GLP.

Acknowledgments

We sincerely thank Mr. Wang from the *Ganoderma* Research Group for assisting in collecting relevant literature and participating in the review and revision of the manuscript. We would also like to express my heartfelt gratitude to the two anonymous peer reviewers for their comprehensive evaluation of the manuscript.

Conflict of Interest Disclosure

The authors affirm that this research was conducted without any commercial or financial relationships that could be construed as a potential conflict of interest.

Reference

- Ahmad F., Ahmad F., Khan M., Alsayegh A., Wahab S., Alam M., and Ahmed F., 2021, *Ganoderma lucidum*: A potential source to surmount viral infections through β -glucans immunomodulatory and triterpenoids antiviral properties, *International Journal of Biological Macromolecules*, 187: 645-656.
<https://doi.org/10.1016/j.ijbiomac.2021.06.122>
- Chang C.J., Lin C.S., Lu C.C., Martel J., Ko Y.F., Ojcius D.M., Tseng S.F., Wu T.R., Chen Y.Y., Young J.D., and Lai H.C., 2015, *Ganoderma lucidum* reduces obesity in mice by modulating the composition of the gut microbiota, *Nature Communications*, 6: 7489.
<https://doi.org/10.1038/ncomms8489>
- Chattopadhyay I., Dhar R., Pethusamy K., Seethy A., Srivastava T., Sah R., Sharma J., and Karmakar S., 2021, Exploring the role of gut microbiome in colon cancer, *Applied Biochemistry and Biotechnology*. 193(6): 1780-1799.
<https://doi.org/10.1007/s12010-021-03498-9>
- Chen M.Y., Xiao D.S., Liu W.W., Song Y.X., Zou B., Li L.H., Li P.P., Cai Y.Y., Liu D.L., Liao Q.J., and Xie Z.X., 2019, Intake of *Ganoderma lucidum* polysaccharides reverses the disturbed gut microbiota and metabolism in type 2 diabetic rats, *International Journal of Biological Macromolecules*, 155: 890-902.
<https://doi.org/10.1016/j.ijbiomac.2019.11.047>
- Gao X.D., Qi J., Ho C.T., Li B., Mu J.Y., Zhang Y.X., Hu H.X., Mo W.C., Chen Z.X., and Xie Y.S., 2020, Structural characterization and immunomodulatory activity of a water-soluble polysaccharide from *Ganoderma leucocontextum* fruiting bodies, *Carbohydrate Polymers*, 249: 116874.
<https://doi.org/10.1016/j.carbpol.2020.116874>
- Guo C.Y., Guo D.L., Fang L.J., Sang T.T., Wu J.W., Guo C.F., Wang Y.Y., Wang Y.L., Chen C.R., Chen J.Q., Chen R.S., and Wang X.L., 2021, *Ganoderma lucidum* polysaccharide modulates gut microbiota and immune cell function to inhibit inflammation and tumorigenesis in colon, *Carbohydrate Polymers*, 267: 118231.
<https://doi.org/10.1016/j.carbpol.2021.118231>
- Guo W.J., Guo J.H., Liu B.L., Lu J.Q., Chen M.Y., Liu B.X., Bai W.J., Rao P.F., Ni L., and Lv X.C., 2020, Ganoderic acid A from *Ganoderma lucidum* ameliorates lipid metabolism and alters gut microbiota composition in hyperlipidemic mice fed a high-fat diet, *Food & Function*, 11(8): 6818-6833.
<https://doi.org/10.1039/d0fo00436g>

- Jin M.L., Zhang H.B., Wang J., Shao D.C., Yang H.Y., Huang Q.L., Shi J.X., Xu C.R., and Zhao K., 2019, Response of intestinal metabolome to polysaccharides from mycelia of *Ganoderma lucidum*, *International Journal of Biological Macromolecules*, 122: 723-731.
<https://doi.org/10.1016/j.ijbiomac.2018.10.224>
- Jin M.L., Zhu Y.H., Shao D.C., Zhao K., Xu C.R., Li Q.H., Yang H.Y., Huang Q.L., and Shi J.X., 2017, Effects of polysaccharide from mycelia of *Ganoderma lucidum* on intestinal barrier functions of rats, *International Journal of Biological Macromolecules*, 94(1): 1-9.
<https://doi.org/10.1016/j.ijbiomac.2016.09.099>
- Kang Q., Chen S.H., Li S.T., Wang B., Liu X.Y., Hao L.L., and Lu J.Q., 2019, Comparison on characterization and antioxidant activity of polysaccharides from *Ganoderma lucidum* by ultrasound and conventional extraction, *International Journal of Biological Macromolecules*, 124: 1137-1144.
<https://doi.org/10.1016/j.ijbiomac.2018.11.215>
- Kendong S., Ali R., Nawawi K., Ahmad H., and Mokhtar N., 2021, Gut dysbiosis and intestinal barrier dysfunction: Potential explanation for early-onset colorectal cancer, *Frontiers in Cellular and Infection Microbiology*, 11: 744606.
<https://doi.org/10.3389/fcimb.2021.744606>
- Khan I., Huang G.Q., Li X.N., Liao W.B., Leong W.F., Xia W.X., Bian X.L., Wu J.M., and Hsiao W.W.L., 2019, Mushroom polysaccharides and jiaogulan saponins exert cancer preventive effects by shaping the gut microbiota and microenvironment in ApcMin/+ mice, *Pharmacological Research*, 148: 104448.
<https://doi.org/10.1016/j.phrs.2019.104448>
- Kubota A., Kobayashi M., Sarashina S., Takeno R., Okamoto K., Narumi K., Furugen A., Suzuki Y., Takahashi N., and Iseki K., 2018, Reishi mushroom *Ganoderma lucidum* modulates IgA production and alpha-defensin expression in the rat small intestine, *Journal of Ethnopharmacology*, 214: 240-243.
<https://doi.org/10.1016/j.jep.2017.12.010>
- Krumbeck J.A., Rasmussen H.E., Hutkins R.W., Clarke J., Shawron K., Keshavarzian A., and Walter J., 2018, Probiotic Bifidobacterium strains and galactooligosaccharides improve intestinal barrier function in obese adults but show no synergism when used together as synbiotics, *Microbiome*, 6(1): 121.
<https://doi.org/10.1186/s40168-018-0494-4>
- Li D.D., Gao L., Li M.M., Luo Y.L., Xie Y.Y., Luo T.T., Su L.F., Yong T.C., Chen S.J., Jiao C.L., Su J.Y., and Huang S.H., 2020, Polysaccharide from spore of *Ganoderma lucidum* ameliorates paclitaxel-induced intestinal barrier injury: Apoptosis inhibition by reversing microtubule polymerization, *Biomedicine & Pharmacotherapy*, 130: 110539.
<https://doi.org/10.1016/j.biopha.2020.110539>
- Li J., Gu F.J., Cai C.Y., Hu M.Y., Fan L.L., Hao J.Y., and Yu G.X., 2019, Purification, structural characterization, and immunomodulatory activity of the polysaccharides from *Ganoderma lucidum*, *International Journal of Biological Macromolecules*, 143: 806-813.
<https://doi.org/10.1016/j.ijbiomac.2019.09.141>
- Li L.X., Liu H.Y., Zhang Q.Y., Li Z.Y., Wong T.F., Fung H.K., Zhang J.R., Bai S.L., Lu A.P., and Han Q.J., 2018, Comprehensive comparison of polysaccharides from *Ganoderma lucidum* and *G. sinense*: Chemical, antitumor, immunomodulating and gut-microbiota modulatory properties, *Scientific Reports*, 8: 22885.
<https://doi.org/10.1038/s41598-018-22885-7>
- Li M.M., Yu L., Zhai Q.X., Chu C.L., Wang S.M., Zhao J.X., Zhang H., Tian F.W., and Chen W., 2023, Combined *Ganoderma lucidum* polysaccharide and ciprofloxacin therapy alleviates *Salmonella enterica* infection, protects the intestinal barrier, and regulates gut microbiota, *Food & Function*, 14(8): 3685-3698.
<https://doi.org/10.1039/d3fo00625e>
- Liu G.Y., Zhang J.X., Hou T.T., An S.Q., Guo B.L., Liu C.F., Hu L.P., Huang Y.B., Zhang S.L., Song M., and Cao Y.L., 2021, Extraction kinetics, physicochemical properties and immunomodulatory activity of the novel continuous phase transition extraction of polysaccharides from *Ganoderma lucidum*, *Food & Function*, 12(20): 9708-9718.
<https://doi.org/10.1039/d1fo02185k>
- Lu J.Q., He R.L., Sun P.H., Zhang F.F., Linhardt R.J., and Zhang A.Q., 2020, Molecular mechanisms of bioactive polysaccharides from *Ganoderma lucidum* (Lingzhi): A review, *International Journal of Biological Macromolecules*, 150: 765-774.
<https://doi.org/10.1016/j.ijbiomac.2020.02.035>
- Luo J.Y., Zhang C.X., Liu R.R., Gao L., Ou S.Y., Liu L.L., and Peng X.C., 2018, *Ganoderma lucidum* polysaccharide alleviating colorectal cancer by alteration of special gut bacteria and regulation of gene expression of colonic epithelial cells, *Journal of Functional Foods*, 47: 127-135.
<https://doi.org/10.1016/j.jff.2018.05.041>
- Lv X.C., Guo W.J., Li L.H., Yu X.H., and Liu B.L., 2019, Polysaccharide peptides from *Ganoderma lucidum* ameliorate lipid metabolic disorders and gut microbiota dysbiosis in high-fat diet-fed rats, *Journal of Functional Foods*, 57: 229-240.
<https://doi.org/10.1016/j.jff.2019.03.043>